



Technical Service Bulletin

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New Fuel Pump Supply Hoses and Electronic Control Valve Assembly

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK23 CM500
- QSK23 CM2250 K109

Issue

Fuel leaks have been observed at bellow location of the tube between fuel filter head and fuel pump.

Symptom:

- Fuel leak from fuel pump to fuel pump and filter head.
- Part appearance may show shift in center of fuel pump supply hose end with respect to elbow fitted at filter head location.

Root Cause:

- Incorrect assembly process can induce stress in the fuel pump supply hose.
- Small crack can develop at beginning or end of bellows section of tube, allowing fuel to leak.

Verification/Confirmation

Key on, run engine at high idle for 5 minutes, allow engine to shutdown properly, key off and inspect fuel pump supply hose from fuel pump to fuel filter head at the beginning and end of bellows section for leaks.

Resolution

Replace existing fuel pump supply hose using Table 1 below. Reference procedures 006-024 and 005-016 in the corresponding Service Manual.

Part Identification

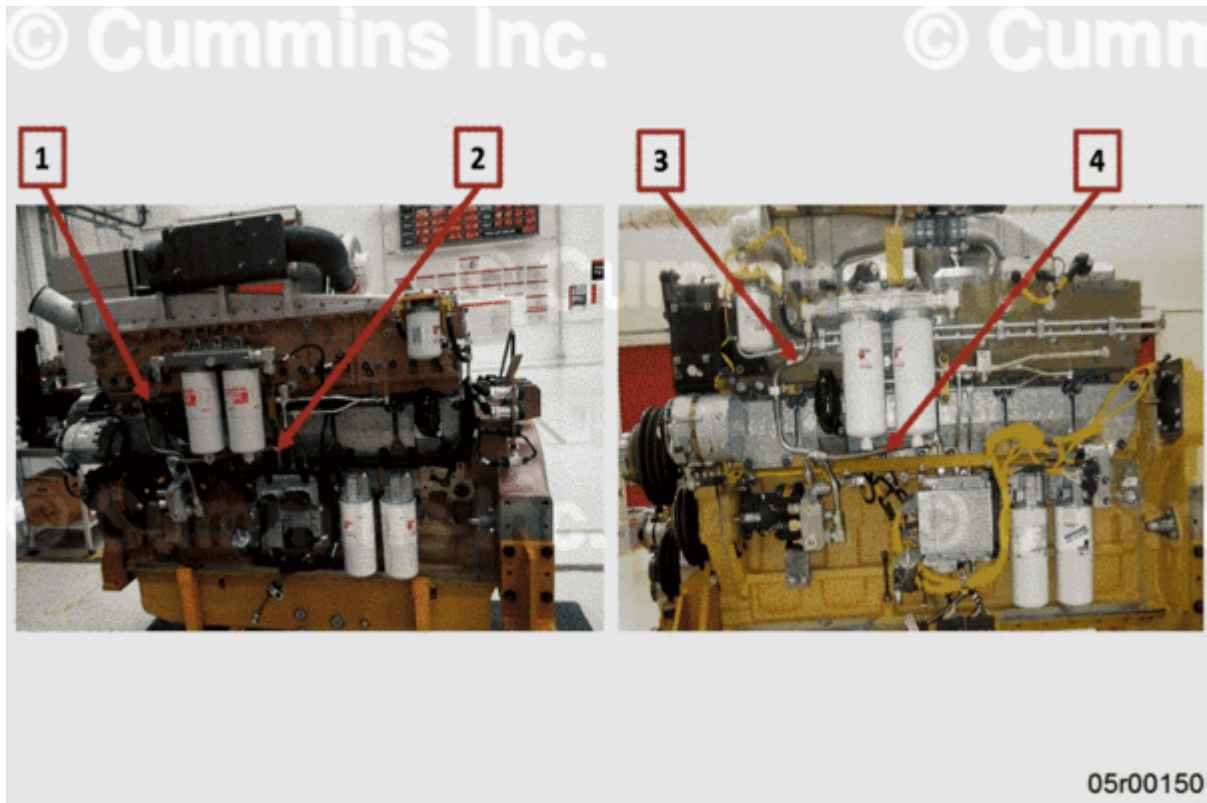


Figure 1, Existing Fuel Hose Arrangement. Left: Power-Gen Application, Right: Industrial Application

Existing fuel pump supply hose Part Numbers in Figure 1:

1. 4095637
2. 4096470
3. 4095774
4. 4096470

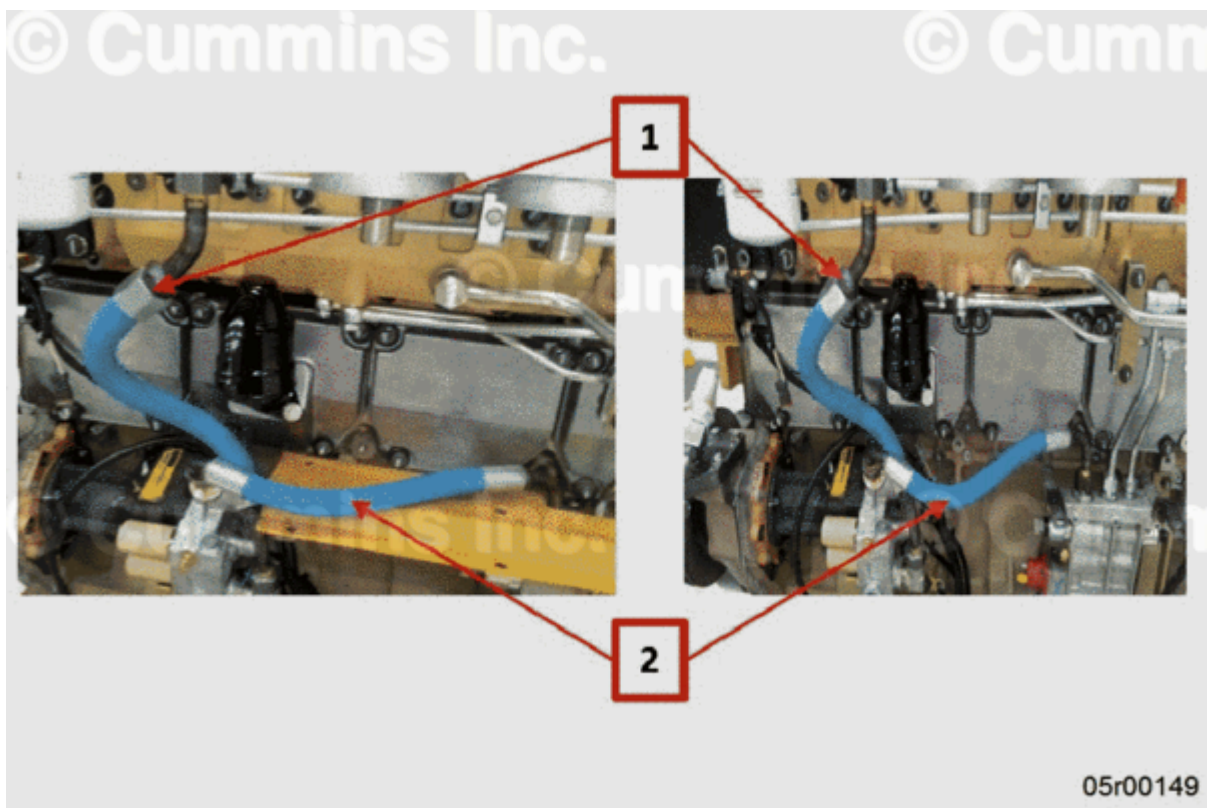


Figure 2, New Fuel Hose Arrangement. Left: Power-Gen Application, Right: Industrial Application

New fuel pump supply hose Part Numbers in Figure 2:

1. 4348438
2. 4399136

Service Instructions

Table 1, Service Instructions

Sequence	Task	Specification
1	Apply oil on the O ring of elbow	-
2	Thread elbow on fuel filter head hand tight	-
3	Thread elbow on fuel pump hand tight	-
4	Thread fuel hose end connector on the elbow of fuel filter head and thread opposite end on the elbow of fuel pump	-
5	Adjust elbow position as per hose alignment & maintain clearance surrounding the fuel hose by adjusting the end connector.	-

Table 1, Service Instructions		
Sequence	Task	Specification
6	Hand tighten with engagement between 60 - 70% of threads of fuel hose nut by hand	-
7	Tighten elbow of fuel filter head at direction of fuel hose end connector using torque wrench and an additional spanner wrench for holding the elbow. Repeat process for fuel pump side.	225 Nm
8	While tightening elbow position should not be changed.	-
9	Tighten fuel hose nut in steps. (Do not apply full torque in one stroke or it could damage the connector end threads)	90 Nm- pump 90 Nm- filter head 68 Nm – pump to ECVA
10	Support the mating fitting with a wrench and tighten the fuel hose nuts using a torque wrench.	-
11	While tightening fuel hose don't allow hose to change position.	-

Service Parts Availability

Service parts are available. See Table 1 for part numbers.

Table 2, Service Parts						
Part Description	Existing Part Number	Obsolete	Superseded	Application	New Part Description	New Part Number
Tube, Fuel Transfer/ Fuel Supply Hose	4095774	Yes	Yes	Industrial	Hose, Fuel	4348438
	4095637	Yes	Yes	Power-Gen		4348438
	4096470	Yes	Yes	Industrial or Power-Gen		4399136

Production Status

Table 3, Production Information

ESN First	Build Date ¹	Plant
85003958	15 December 2017	Phaltan Engine Plant
¹ Engine build date can be found on engine data plate.		

Publications Affected

Table 4, Publications Affected

Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	QSK23 Industrial	4021375	Fuel Supply Lines	006-024	6
Service Manual	QSK23 P-Gen	4367358	Fuel Supply Lines	006-024	6

Document History

Date	Details
2018-6-13	Module Created

Last Modified: 05-Jul-2018
